The three School scaling FFA rings

Field model, magnet placement and space charge

A Adelmann

13 November 2026

The School contains three fixed-energy proton lattices. Two repeat one F/D cell sixteen times at 12 MeV. The third repeats four unequal F/D pairs four times at 3 MeV. All lengths below are in metres, fields in tesla and angles in radians unless degrees are shown explicitly.

Table 1: Parameters of the three inputs. R_0 is the geometric reference radius, not the closed-orbit launch radius.

Lattice	T [MeV]	R_0 [m]	k	δ	Periods
16-cell F/D	12	4.42	7.7170220643	45.005835°	16
55°, D off	12	4.42	7.7170220643	55°	16
Fourfold	3	4.00	7.4319	30°	4

The fourth directory, `school-16-cell-probe`, adds diagnostic probes to the first magnetic lattice. It is not a fourth machine. Its beam launch and tracking settings may differ; its geometry export still describes the same magnet layout.

1 Coordinates and absolute placement

The ring lies in the global X–Z plane, with Y vertical. The launch origin lies near the beam, whose initial forward direction is $+Z$. The reference circle has centre $\boldsymbol{C} = (-R_0, 0, 0)$:

$$\boldsymbol{R}(\varphi) = \begin{pmatrix} R_0(\cos\varphi - 1) \\ 0 \\ R_0 \sin\varphi \end{pmatrix}, \quad s = R_0\varphi, \quad \boldsymbol{t}(\varphi) = \begin{pmatrix} -\sin\varphi \\ 0 \\ \cos\varphi \end{pmatrix}. \tag{1}$$

Here s is reference arc length and $\boldsymbol{t}$ the unit tangent. A magnet with entrance azimuth α_i uses the explicit OPALX pose

$$\boxed{\begin{aligned} X_i &= R_0(\cos\alpha_i - 1), \quad & Y_i &= 0, \quad & Z_i &= R_0 \sin\alpha_i, \\ \texttt{THETA}_i &= -\alpha_i, & \texttt{PHI}_i &= 0, \quad & \texttt{PSI}_i &= 0. \end{aligned}} \tag{2}$$

The position denotes the *entrance*, not the body centre. Local x points radially outwards and local z follows the entrance tangent. The transformation from local coordinates to the laboratory is

$$\boldsymbol{r} = \boldsymbol{R}(\alpha_i) + Q_i \boldsymbol{r}_i, \qquad Q_i = \begin{pmatrix} \cos\alpha_i & 0 & -\sin\alpha_i \\ 0 & 1 & 0 \\ \sin\alpha_i & 0 & \cos\alpha_i \end{pmatrix}. \tag{3}$$

Consequently, $Q_i(0,0,1)^{\mathsf{T}} = \boldsymbol{t}(\alpha_i)$ and $\boldsymbol{R}(\alpha_i) + Q_i(-R_0, 0, 0)^{\mathsf{T}} = \boldsymbol{C}$. Every element therefore shares the same curvature centre. A body of arc length ℓ_i exits at $\boldsymbol{R}(\alpha_i + \ell_i/R_0)$.

The fringe-model centre is halfway along the nominal body:

$$\varphi_{c,i} = \alpha_i + \frac{\ell_i}{2R_0}. \tag{4}$$

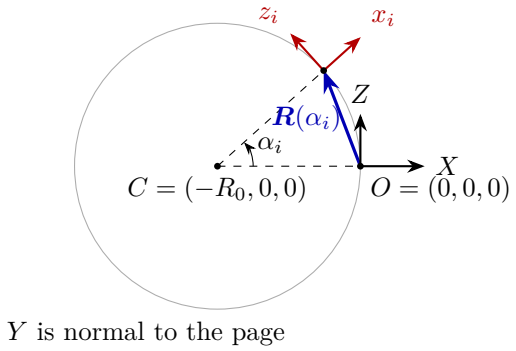


Figure 1: Entrance placement in the horizontal plane. The blue vector gives the translation from O to the entrance. The red local axes are radial and tangential. The reference circle is a geometry construction, not a tracked particle orbit.

2 Spiral faces and fringe fields

For positive R_0, the common cylindrical and spiral coordinates are

$$\begin{aligned} r &= \sqrt{(X + R_0)^2 + Z^2}, \qquad \varphi = \operatorname{atan2}(Z, X + R_0), \\ \chi_i &= \varphi - \varphi_{c,i} - \tan\delta \ln(r/R_0). \end{aligned} \tag{5}$$

The azimuth uses the branch local to the element. Constant χ_i describes a logarithmic spiral. In particular, the nominal body faces satisfy

$$\boxed{\begin{aligned} \varphi_{i,\mathrm{in}}(r) &= \alpha_i + \tan\delta \ln(r/R_0), \\ \varphi_{i,\mathrm{out}}(r) &= \alpha_i + \ell_i/R_0 + \tan\delta \ln(r/R_0). \end{aligned}} \tag{6}$$

Since $r\,\mathrm{d}\varphi/\mathrm{d}r = \tan\delta$, a face makes the constant angle δ with the radial direction. Changing δ changes the face shape, while leaving its entrance at $r = R_0$ fixed.

The midplane scaling law separates radial strength and azimuthal fringe shape:

$$B_Y(r, 0, \varphi) = \left(\frac{r}{R_0}\right)^k \sum_i B_{0i} F_i(R_0 \chi_i). \tag{7}$$

The sum includes elements whose field support contains the evaluation point. The School uses symmetric parameters within an `ASYMMETRIC_ENGE` model:

$$E(u) = \frac{1}{1 + \exp[\sum_{n=0}^{n_a} a_n (u/\lambda)^n]}, \qquad F_i(s) = E(-s - \ell_i/2) + E(s - \ell_i/2) - 1. \tag{8}$$

Here λ is the fringe-length parameter and ℓ_i the fringe model's centre length. Neither is an integrated magnetic length. The nominal faces are not exact half-field contours for a general Enge polynomial; fields extend beyond them. The vertical field expansion retains `MAX_Y_POWER=4`.

The aperture is $R_0 - \Delta r_- \le r \le R_0 + \Delta r_+$, $|Y| \le H/2$. The input `AZIMUTHAL_EXTENT` is a length A; its angular support is $|\chi_i| \le A/R_0$. All three machines use $A = R_0(2\pi/16)$. This window is wider than the body and permits overlapping fringe contributions in Eq. (7).

3 Sixteen-cell F/D ring at 12 MeV

The reference circumference is $2\pi R_0 = 27.771679$ m. Let $\Delta = 2\pi/16 = 22.5°$. The sixteen cells have entrance azimuths

$$\boxed{\alpha_{j,F} = 6.5425° + j\Delta, \qquad \alpha_{j,D} = 13.40625° + j\Delta.} \tag{9}$$

Here $j = 0, \ldots, 15$. The centre azimuths are $8.7925° + j\Delta$ and $14.53125° + j\Delta$. The lengths are $\ell_F = 0.3471459882$ m and $\ell_D = 0.1735729941$ m.

The OPALX field coefficients are $B_{0F} = +0.6532097983$ T and $B_{0D} = -0.5027652364$ T. The input uses $\delta = 3.142/4$ rad, rather than exactly $\pi/4$. Its Enge coefficients are $(a_0, a_1, a_2, a_3) = (-1.3, 3.42, -1.20, 0.23)$ with $\lambda = 0.12$ m. The aperture parameters are $\Delta r_- = 0.8$ m, $\Delta r_+ = 1.2$ m and $H = 1$ m. Zero-field arcs fill the gaps between bodies and close the reference geometry.

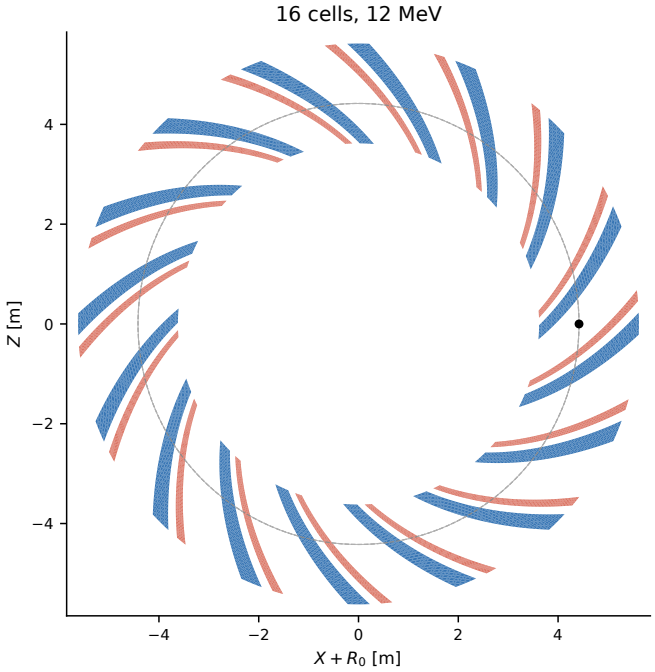


Figure 2: Sixteen-cell F/D ring at 12 MeV. F bodies are blue, D bodies red; zero-field arcs are grey. The dashed circle has radius R_0; the black dot marks the global origin. The horizontal coordinate is translated by R_0 to centre the ring. No particle trajectory is plotted.

3.1 Space charge: from a KV beam to particle tracking

A proton beam repels itself. The electric self-force pushes particles away from the beam centre and reduces the restoring force of the magnets. Consequently, the transverse oscillations usually slow down: their tunes decrease with beam current. This section derives an analytic prediction and compares it with an OPALX simulation of the same 16-cell lattice. The analytic model explains the mechanism; the particle calculation tests the approximations.

Here s denotes arc length along the *magnetic closed orbit*, rather than along the placement circle in Eq. (1). Local x is horizontal and normal to that orbit, and y is vertical. These lower-case coordinates describe displacements from the orbit. A prime denotes differentiation with respect to s. The paraxial approximation assumes small transverse slopes, $|x'|, |y'| \ll 1$. We first assume fixed energy, uncoupled transverse motion and a long beam with negligible longitudinal self-force.

3.1.1 Which current sets the strength of space charge?

For protons of rest mass m and charge $q > 0$, define

$$\gamma = 1 + \frac{T}{mc^2}, \qquad \beta = \frac{v}{c} = \sqrt{1 - \gamma^{-2}}, \qquad I = \lambda \beta c. \tag{10}$$

The line-charge density λ has units C/m. A uniformly populated bunch of full length L_b and total charge $\mathcal{Q}_b$ has

$$\lambda = \frac{\mathcal{Q}_b}{L_b}, \qquad \mathcal{Q}_b = \frac{I L_b}{\beta c}, \qquad \overline{I} = \frac{\mathcal{Q}_b}{T_{\rm rev}} = I \frac{L_b}{C_{\rm co}}, \tag{11}$$

where $C_{\rm co}$ is the closed-orbit length and $T_{\rm rev} = C_{\rm co}/(\beta c)$. The current I is the current *inside the bunch*; $\overline{I}$ is the current averaged over a revolution for one bunch. They are different when most of the ring is empty. Local transverse space charge depends on I.

At 12 MeV, $\beta = 0.15842$, $\gamma = 1.01279$ and $C_{\rm co} \simeq 28.0297$ m. For $L_b = 0.20$ m, a bunch current of 10.399 mA therefore requires only 43.793 pC; its revolution-averaged current is about 74.2 μA. When comparing this short FFA bunch with a longer bunch in a different lattice, we keep the *bunch current* fixed, not the total charge.

3.1.2 What is special about the KV distribution?

The Kapchinskij–Vladimirskij (KV) distribution is an idealized transverse phase-space distribution whose projection into position space is a uniformly charged ellipse.[1] Its usefulness follows from that uniform density: the electric field inside the ellipse is linear in displacement.

For either plane $j = x, y$, the rms emittance and Twiss parameters describe the second moments about the beam centroid:

$$\begin{aligned} \varepsilon_j &= \sqrt{\langle j^2 \rangle \langle j'^2 \rangle - \langle j j' \rangle^2}, \\ \Sigma_j &= \begin{pmatrix} \langle j^2 \rangle & \langle j j' \rangle \\ \langle j j' \rangle & \langle j'^2 \rangle \end{pmatrix} = \varepsilon_j \begin{pmatrix} \beta_j^{\rm Tw} & -\alpha_j^{\rm Tw} \\ -\alpha_j^{\rm Tw} & \gamma_j^{\rm Tw} \end{pmatrix}, \qquad \gamma_j^{\rm Tw} = \frac{1 + (\alpha_j^{\rm Tw})^2}{\beta_j^{\rm Tw}}. \end{aligned} \tag{12}$$

The superscript "Tw" distinguishes the optical functions from the relativistic factors β and γ. The brackets denote an average over particles, with the centroid subtracted. The geometric rms emittance ε_j has units m rad; the normalized rms emittance is $\varepsilon_{n,j} = \beta\gamma\varepsilon_j$ at fixed energy.

For each current, use its matched Twiss parameters and rms emittances to introduce dimensionless normalized coordinates,

$$u_j = \frac{j}{\sqrt{\varepsilon_j \beta_j^{\rm Tw}}}, \qquad v_j = \frac{\alpha_j^{\rm Tw} j + \beta_j^{\rm Tw} j'}{\sqrt{\varepsilon_j \beta_j^{\rm Tw}}}. \tag{13}$$

[1] I. M. Kapchinskij and V. V. Vladimirskij, "Limitations of proton beam current in a strong focusing linear accelerator associated with the beam space charge," in the 1959 CERN conference proceedings, p. 274 ff., especially Sections II–IV; original paper.

A KV beam samples the *surface* of a four-dimensional sphere:

$$f_{\mathrm{KV}} \;\propto\; \delta\Big(u_x^2 + v_x^2 + u_y^2 + v_y^2 - 4\Big)\,. \tag{14}$$

Thus it is not a uniformly filled four-dimensional ball and not a Gaussian. Integrating over the two momentum coordinates gives a uniform position density inside an ellipse. Its edge semi-axes a, b are twice the rms sizes:

$$a = 2\sigma_x, \qquad b = 2\sigma_y, \qquad \rho(x,y) = \frac{\lambda}{\pi a b} \quad \text{for} \quad \frac{x^2}{a^2} + \frac{y^2}{b^2} \le 1, \tag{15}$$

with zero density outside. Mixing edge radii and rms sizes gives incorrect space-charge factors. All emittances in this section are *rms* emittances.

3.1.3 Linear self-forces and the envelope equations

In free space, the transverse electric field inside that ellipse is

$$E_x = \frac{\lambda}{\pi\epsilon_0}\frac{x}{a(a+b)}, \qquad E_y = \frac{\lambda}{\pi\epsilon_0}\frac{y}{b(a+b)}. \tag{16}$$

For a round beam, $a = b$, this reduces to $E_r = \lambda r/(2\pi\epsilon_0 a^2)$, as obtained directly from Gauss's law. The beam current also produces a magnetic self-field. For particles moving approximately together, its inward force partially cancels the electric force:

$$\boldsymbol{B}_{\mathrm{sc}} = \frac{\beta}{c}\,\boldsymbol{t} \times \boldsymbol{E}_\perp, \qquad \boldsymbol{F}_\perp \simeq q(1-\beta^2)\boldsymbol{E}_\perp = \frac{q}{\gamma^2}\boldsymbol{E}_\perp. \tag{17}$$

Here $\boldsymbol{t}$ is the orbit tangent. The electric repulsion dominates, so the net self-force remains defocusing.

The dimensionless generalized perveance collects the current and energy scaling:

$$K = \frac{2I}{I_A\beta^3\gamma^3}, \qquad I_A = \frac{4\pi\epsilon_0 m c^3}{q} \simeq 31.30\times 10^6 \text{ A} \quad \text{for protons.} \tag{18}$$

The proton value must be used here, not the much smaller electron value. Dividing the transverse force by $m\gamma\beta^2c^2$ converts it to an acceleration with respect to s. If $\kappa_x(s)$ and $\kappa_y(s)$ denote the external linear focusing functions, the particle equations become

$$\begin{aligned} x'' + [\kappa_x(s) - g_x(s)]x = 0, \quad g_x &= \frac{2K}{a(a+b)}, \\ y'' + [\kappa_y(s) - g_y(s)]y = 0, \quad g_y &= \frac{2K}{b(a+b)}. \end{aligned} \tag{19}$$

Both κ_j and g_j have units m^{-2}. The positive g_j reduces the external focusing. For a KV distribution, its rms envelopes obey

$$\boxed{\begin{aligned} \sigma_x'' + \kappa_x\sigma_x - \frac{\varepsilon_x^2}{\sigma_x^3} - \frac{K}{2(\sigma_x+\sigma_y)} = 0, \\ \sigma_y'' + \kappa_y\sigma_y - \frac{\varepsilon_y^2}{\sigma_y^3} - \frac{K}{2(\sigma_x+\sigma_y)} = 0. \end{aligned}} \tag{20}$$

The four terms describe envelope curvature, external focusing, emittance pressure and space-charge defocusing. For example, multiplying the horizontal equation by two and substituting $a = 2\sigma_x$ gives an emittance term $16\varepsilon_x^2/a^3$ and a space-charge term $2K/(a+b)$ in the edge-radius equation. This provides a useful check on conventions.

3.1.4 Matching the beam and predicting its tunes

A matched beam breathes with the periodic lattice, but repeats its envelope and envelope slope after one cell. With cell length $\ell_c = C_{\mathrm{co}}/16$, the matching conditions are

$$\sigma_j(s+\ell_c) = \sigma_j(s), \qquad \sigma_j'(s+\ell_c) = \sigma_j'(s). \tag{21}$$

To find a matched beam at a chosen current, keep the two emittances fixed and adjust the initial sizes and correlations until these conditions hold. Reusing the zero-current sizes without rematching would generally excite envelope oscillations and complicate the tune comparison.

Our analytic calculation uses linear transfer maps obtained by differentiating the magnetic equations about the verified closed orbit. It alternates these external maps with short KV defocusing kicks. In slope coordinates a kick of length Δs in either plane is

$$L_j = \begin{pmatrix} 1 & 0 \\ g_j \Delta s & 1 \end{pmatrix}, \qquad \Sigma_j \longmapsto L_j \Sigma_j L_j^{\mathsf{T}}. \tag{22}$$

The actual calculation uses mechanical momenta normalized by mc and transforms the radial reference-section coordinates to the local orbit-normal frame. In that momentum convention $p_x/(mc) \simeq \beta\gamma x'$, so the lower-left kick element includes an additional factor $\beta\gamma$. The physical force is the same. Half kicks on both sides of each external map give a symmetric integration. Iterating the covariance to periodicity implements Eq. (21); this is an analytic-field model solved numerically, not a fit to the OPALX tune measurements.

Let $M_{j,c}$ be the resulting matched one-cell particle matrix in plane j. For stable uncoupled motion, its eigenvalues are $\exp(\pm i\mu_j)$, with

$$\cos\mu_j = \tfrac{1}{2}\,\mathrm{Tr}\, M_{j,c}, \qquad Q_j = \frac{16\mu_j}{2\pi}, \qquad \Delta Q_j(I) = Q_j(I) - Q_j(0). \tag{23}$$

The phase branch is followed continuously from the zero-current solution. The tune counts *particle oscillations per revolution*; it is not the frequency of the centroid or the envelope. In the weak-space-charge limit, first-order optics gives the useful estimate

$$\Delta Q_j \simeq -\frac{1}{4\pi}\oint \beta_{j,0}^{\mathrm{Tw}}(s) g_j(s)\,\mathrm{d}s. \tag{24}$$

The minus sign explains tune depression. The full matched calculation also allows the envelope and therefore g_j to change with current.

3.1.5 What OPALX calculates with FFT2D5

OPALX follows individual macroparticles in the nonlinear magnetic fields introduced in Section 2. A macroparticle represents many protons: with N_{mac} equal-weight particles its deposited charge is $\mathcal{Q}_b/N_{\mathrm{mac}}$, while the tracked charge-to-mass ratio remains the proton value. The distribution can deform during tracking; OPALX does not force it to remain KV or enforce linear self-fields.

The FFT2D5 solver uses a *particle-in-cell* method. It resolves the charge on a three-dimensional mesh, but solves a two-dimensional transverse Poisson problem for each longitudinal slice. This is the meaning of "2.5D." It is a local, long-beam approximation, rather than a full three-dimensional Poisson solution. For this study the sequence is:

1. Project each particle onto the supplied closed reference path and form local transverse coordinates and an arc-length coordinate s.

2. Deposit macroparticle charges with cloud-in-cell weights. For constant timestep and longitudinal cloud-in-cell deposition, the density has the form

$$\rho_{ijk} = \frac{1}{\Delta x \Delta y \Delta s}\sum_p q_p W_i(x_p) W_j(y_p) W_k(s_p), \tag{25}$$

where each linear weight W connects neighbouring cell centres and the weights for a particle sum to one.

3. In each occupied slice k, solve

$$\left(\partial_x^2 + \partial_y^2\right)\Phi_k = -\frac{\rho_k}{\epsilon_0}, \qquad \boldsymbol{E}_{\perp,k} = -\nabla_\perp \Phi_k. \tag{26}$$

The solver evaluates the free-space logarithmic Green-function convolution with fast Fourier transforms (FFTs). It doubles the transverse computational domain for the convolution and integrates the Green function over a mesh cell, including the singular self-cell. Finite differences of the potential give the electric field. FFTs accelerate this calculation; they do not impose a conducting transverse wall.

4. Interpolate the field back to particles with the corresponding cloud-in-cell weights. Construct the transverse magnetic self-field as in Eq. (17) and rotate both fields back to the laboratory. The transverse Poisson source already uses charge per *laboratory* length, so its electric field is already the laboratory field; applying another factor γ would double-count a transformation.

5. Add the external magnetic fields and advance the relativistic Lorentz equations with the Boris integrator:

$$\frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t} = \frac{\boldsymbol{p}}{m\gamma_p}, \qquad \frac{\mathrm{d}\boldsymbol{p}}{\mathrm{d}t} = q\left[\boldsymbol{E}_{\mathrm{sc}} + \boldsymbol{v} \times (\boldsymbol{B}_{\mathrm{ext}} + \boldsymbol{B}_{\mathrm{sc}})\right], \quad \gamma_p = \sqrt{1 + \frac{|\boldsymbol{p}|^2}{m^2c^2}}. \tag{27}$$

The setting `SCFIELDUPDATE=MIDPOINT` evaluates the collective field at the midpoint stage of the time step.

In this scan, `PIPEMODE=NONE` disables the approximate longitudinal space-charge field: $E_s^{\mathrm{sc}} = 0$. Transverse space charge remains active. The bunch ends therefore do not generate longitudinal self-forces in this model. The parameters `PIPESIZEX` and `PIPESIZEY` set the transverse mesh widths; they do not specify conducting-wall boundary conditions for this transverse free-space solve. With `CLOSEDRING=TRUE`, the longitudinal coordinate wraps around the closed reference path.

3.1.6 The school scan and its numerical resolution

The zero-current launch has radial rms size 5.0 mm and vertical rms size 2.4 mm at the $Z = 0$ reference section. The horizontal size here is a radial section measurement; the KV self-force uses the size after transformation to the local orbit-normal plane. The geometric rms emittances are approximately 26.338 and 3.748 mm mrad, or normalized rms emittances 4.226 and 0.6014 mm mrad. Matching at each nonzero current changes the sizes slightly while keeping the emittances fixed. Each 20 cm bunch contains 98,304 macroparticles. The source uses 128 uniformly spaced longitudinal midpoint slices and a symmetric, low-noise sample of the KV shell in each slice. These source slices and the field-mesh slices are distinct discretizations.

The transverse field grid has 64×64 cells across 64×64 mm, so $\Delta x = \Delta y = 1$ mm. The longitudinal grid spans the *entire* reference path, including the empty part of the ring. Its spacing and the number of cells occupied by the bunch are approximately

$$\Delta s = \frac{C_{\mathrm{co}}}{N_s}, \qquad N_{s,\mathrm{bunch}} = \frac{L_b}{\Delta s} = \frac{L_b N_s}{C_{\mathrm{co}}}. \tag{28}$$

Choosing $N_s = 17920$ gives $\Delta s = 1.564$ mm and about 128 cells across 20 cm. Setting the full-ring value to $N_s = 128$ would instead give $\Delta s = 21.9$ cm, leaving less than one cell across the bunch. That would smear the line density and underestimate the local self-force.

The time step is 50 ps. Short pilot comparisons checked particle IDs, finite coordinates, charge, one versus two GPU ranks, and 50 versus 25 ps steps. The latter comparison gives a maximum position difference of 0.276 μm after 25 ns. These are useful implementation checks, but do not establish long-time tune convergence. The target record is 128 turns; the preliminary comparison below uses the same first 15 turns in every case.

Table 2: School intensity scan. The current is the current inside the uniform 20 cm bunch. The KV tunes come from the periodic linear self-field model.

I [mA]	$\mathcal{Q}_b$ [pC]	$Q_{x,\mathrm{KV}}$	$Q_{y,\mathrm{KV}}$
0	0	3.410660	3.391112
4.749	19.998	3.408404	3.385117
8.701	36.643	3.406527	3.380131
10.399	43.793	3.405722	3.377991

3.1.7 Reading the tune footprint and its colour bar

A tune footprint places one point (Q_x, Q_y) for each analysed particle. We estimate the tunes with Numerical Analysis of Fundamental Frequencies (NAFF), using position records resampled at equal azimuth intervals and a Hann window of order two. The horizontal closed-orbit scalloping is removed before interpolation; the central-slice centroid is removed to measure incoherent particle tunes. All four cases use 768 corresponding particles from the central longitudinal slice and the same 15-turn interval, starting one-quarter turn after launch. Corresponding particles have the same normalized launch directions; matching changes their physical coordinates slightly as the current increases.

The colour identifies how each particle initially shares its *normalized* transverse action between the two planes. Define the Courant–Snyder action and its dimensionless version by

$$\mathcal{I}_j = \frac{\gamma_j^{\mathrm{Tw}} j^2 + 2\alpha_j^{\mathrm{Tw}} j j' + \beta_j^{\mathrm{Tw}} j'^2}{2}, \qquad \frac{\mathcal{I}_j}{\varepsilon_j} = \frac{u_j^2 + v_j^2}{2}. \tag{29}$$

The KV shell fixes the sum of the normalized actions to two. The horizontal fraction used by the colour bar is therefore

$$\boxed{f_x = \frac{\mathcal{I}_x/\varepsilon_x}{\mathcal{I}_x/\varepsilon_x + \mathcal{I}_y/\varepsilon_y} = \frac{u_x^2 + v_x^2}{4}, \qquad 0 \le f_x \le 1.} \tag{30}$$

Dark purple, near $f_x = 0$, denotes particles with predominantly vertical normalized oscillation amplitude. Yellow, near $f_x = 1$, denotes predominantly horizontal normalized amplitude. At $f_x = 0.5$, the two normalized actions are equal. The corresponding ratios can be read directly:

$$\frac{\mathcal{I}_x}{\varepsilon_x} = 2f_x, \qquad \frac{\mathcal{I}_y}{\varepsilon_y} = 2(1 - f_x). \tag{31}$$

For example, $f_x = 0.75$ means normalized actions 1.5 horizontally and 0.5 vertically: a ratio of three to one. It does *not* mean that the horizontal position is three times the vertical position. Position also depends on oscillation phase and optical functions. Because the two emittances differ, equal normalized actions do not mean equal actions in physical units, equal amplitudes in millimetres, or equal transverse kinetic energies.

The colours are *initial labels*. The same label is used for corresponding particles at every current. In the ideal matched linear KV model the actions are invariants. In the nonlinear FFA fields, these initial linear actions need not stay constant. A colour trend across a footprint shows the dependence of the measured tunes on the initial action sharing; it is not a time sequence, a

current scale, or a density scale. The red crosses mark analytic KV tunes and are not assigned a value on the colour bar. The linear KV model predicts one tune pair for the matched core, whereas finite-amplitude magnetic motion and a nonideal self-field can produce a footprint.

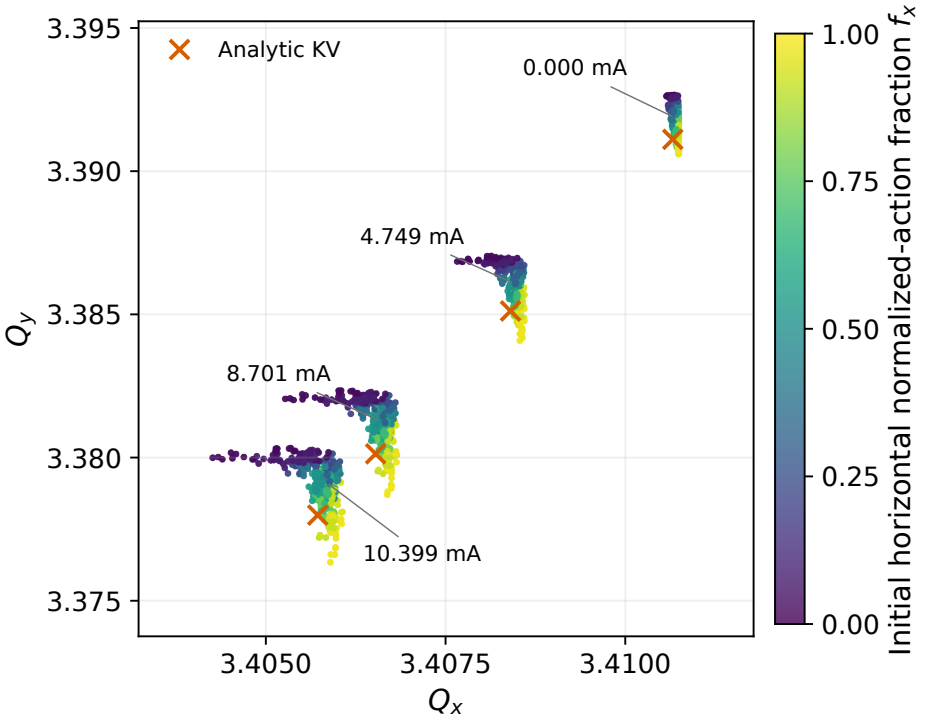


Figure 3: Preliminary FFA tune footprints at bunch currents 0, 4.749, 8.701 and 10.399 mA, labelled next to their particle clouds. Each cloud contains NAFF tunes from 768 central-slice particles over the same 15 turns; red crosses are the corresponding analytic KV predictions. The colour bar gives the *initial horizontal normalized-action fraction* $f_x = (u_x^2 + v_x^2)/4$: purple means mainly vertical action, yellow mainly horizontal action, and $f_x = 0.5$ equal normalized actions. It does not encode current or charge. The four clouds shift towards smaller tunes with current; the zero-current cloud already has a finite-amplitude spread.

3.1.8 Comparing the tune shifts fairly

To isolate the intensity dependence, subtract each method's own zero-current baseline:

$$\begin{aligned} \Delta Q_{j,\mathrm{NAFF}}(I) &= \mathrm{median}_p\, Q_{j,p}(I) - \mathrm{median}_p\, Q_{j,p}(0), \\ \Delta Q_{j,\mathrm{KV}}(I) &= Q_{j,\mathrm{KV}}(I) - Q_{j,\mathrm{KV}}(0). \end{aligned} \tag{32}$$

This definition is a difference of medians, which need not equal the median of the per-particle differences. It also removes the offset between the finite-amplitude zero-current particle median and the linear zero-amplitude KV prediction. That offset is about 7.35×10^{-4} vertically in the present short record.

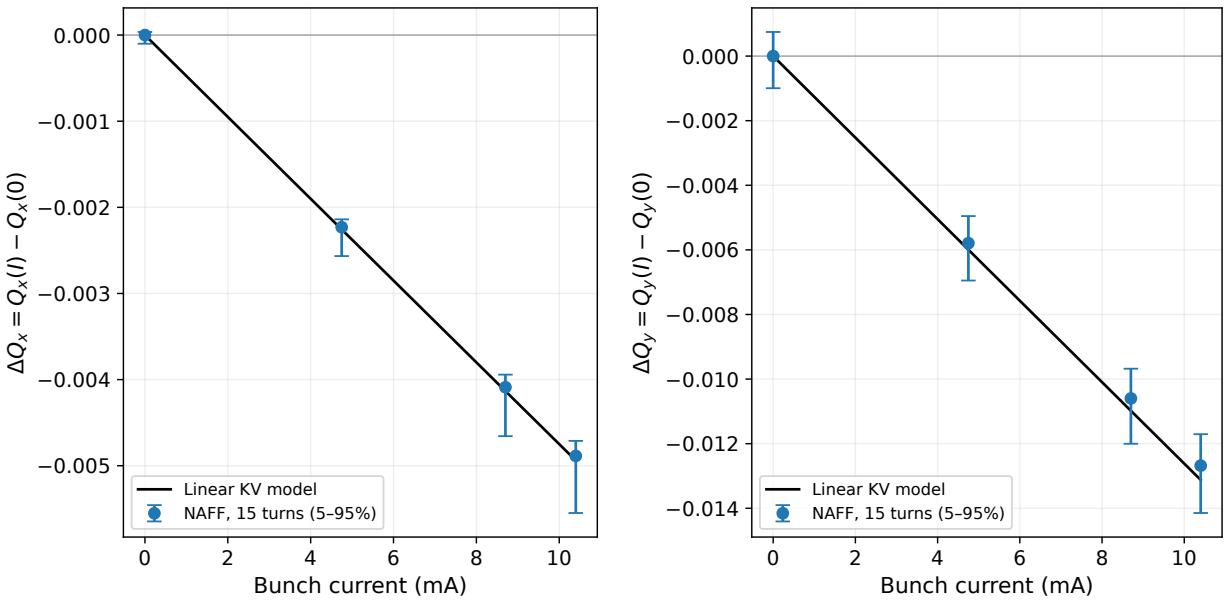


Figure 4: Preliminary tune shifts relative to each method's own zero-current baseline, for the same 15-turn records and central-slice particles as Fig. 3. Dots are differences of NAFF median tunes; black curves connect the matched linear KV predictions. Bars give the 5th–95th percentile particle footprint at each current, shifted by the zero-current median. They are *not* uncertainty bars on the median or a distribution of paired-particle tune differences. The NAFF and KV results show similar tune depression in both planes.

At 10.399 mA the median-based shifts are approximately $\Delta Q_x = -0.004886$ and $\Delta Q_y = -0.012680$, compared with KV values -0.004939 and -0.013122. The simulated depressions are about 1.1% and 3.4% smaller in magnitude. This is an encouraging preliminary trend, not a percent-level accuracy result. Changing the NAFF record from 11 to 15 turns changes individual tunes by up to 7.6×10^{-5} horizontally and 3.9×10^{-4} vertically at the 95th percentile. A short-window estimate can move when several stationary spectral lines overlap; that alone does not demonstrate a changing fundamental tune or chaotic diffusion.

Before assigning the residual difference to a physical mechanism, compare longer records, check the tune footprint against record length, and test particle number, field-grid resolution and timestep. The KV model assumes linear external focusing and a uniform elliptical density with no longitudinal self-force. OPALX retains nonlinear external fields and evolves a finite macroparticle distribution on a mesh, while using the same longitudinal-force approximation here. These distinctions explain what the comparison tests, without identifying a cause for the remaining discrepancy.

4 The 55° ring with D magnets off

The second machine keeps the first ring's radius, energy, field index, Enge coefficients, apertures, body lengths and absolute entrance poses. It changes only the spiral angle and D strength:

$$\delta = 55^\circ, \qquad B_{0F} = +0.6532097983 \text{ T}, \qquad B_{0D} = 0. \tag{33}$$

Sixteen F magnets remain active. The sixteen D elements retain their nominal arcs and apertures but contribute no field. Turning them off is therefore distinct from deleting their geometry or shifting the F magnets. The larger spiral angle changes each F face's radial sweep through Eq. (6).

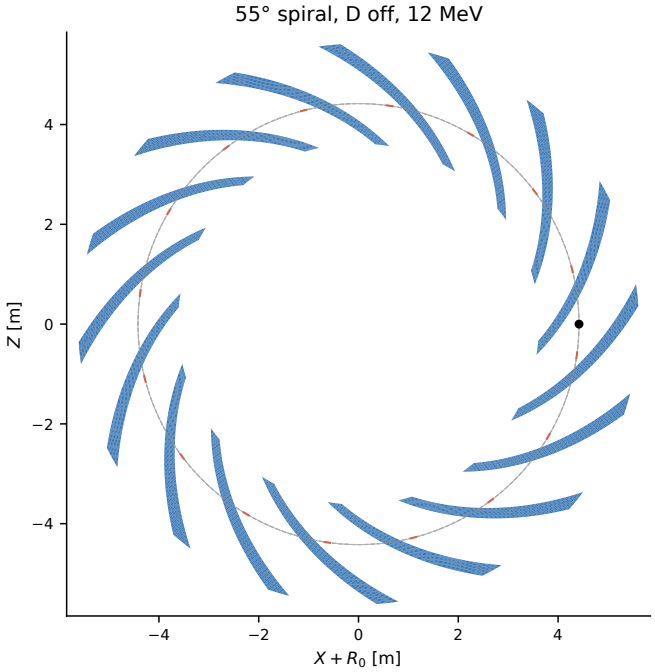


Figure 5: The 55° ring with D magnets off. Active F bodies are blue; dashed red arcs mark the zero-field D elements. The dashed circle has radius R_0; the black dot marks the global origin. The horizontal coordinate is translated by R_0 to centre the ring. No particle trajectory is plotted.

5 Fourfold ring at 3 MeV

The reference circumference is $2\pi R_0 = 25.132741$ m. Sixteen F/D pairs form four repetitions of a four-pair superperiod. The pairs differ in both strength and spacing; the complete ring has fourfold, not sixteenfold, symmetry. The input's `N_CELLS=16` specifies the nominal pair count.

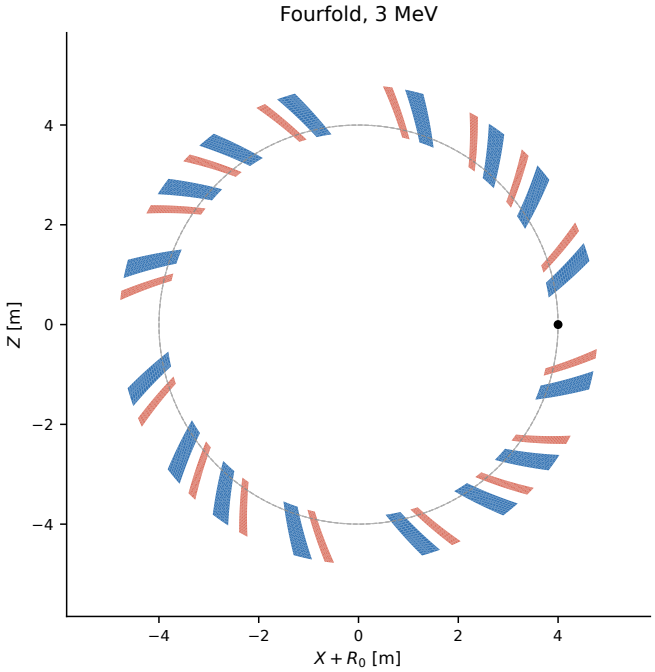


Figure 6: Fourfold ring at 3 MeV. F bodies are blue, D bodies red; zero-field arcs are grey. The dashed circle has radius R_0; the black dot marks the global origin. The horizontal coordinate is translated by R_0 to centre the ring. No particle trajectory is plotted.

For $\Delta = 2\pi/16$, the angular spacing corrections are

$$(\epsilon_1, \epsilon_2, \epsilon_3, \epsilon_4) = (-0.0125, -0.125, -0.0125, +0.15). \tag{34}$$

These numbers are in *radians*. Each pair occupies the reference arc $R_0(\Delta + \epsilon_j)$, while its body lengths remain

$$\ell_F = 0.2\Delta R_0 = 0.3141592654 \text{ m}, \qquad \ell_D = 0.1\Delta R_0 = 0.1570796327 \text{ m}. \tag{35}$$

Define the angular pair starts within a superperiod by

$$q_1 = 0, \qquad q_j = \sum_{h=1}^{j-1} (\Delta + \epsilon_h). \tag{36}$$

Since $\sum_j \epsilon_j = 0$, four pairs span exactly $4\Delta = \pi/2$. For superperiod $m = 0, 1, 2, 3$, set $\Phi_m = \Phi_0 + m\pi/2$ with $\Phi_0 = (0.6\Delta + 0.15)/2 = 0.1928097245$ rad. The entrance angles are

$$\boxed{\begin{aligned} \alpha_{m,j,F} &= \Phi_m + q_j - 0.075\Delta, \\ \alpha_{m,j,D} &= \Phi_m + q_j + 0.2625\Delta. \end{aligned}} \tag{37}$$

Combining Eqs. (2) and (37) specifies every entrance translation and rotation. The first-superperiod values are listed below.

Table 3: Fourfold superperiod. These are OPALX B_0 values; all rows repeat at azimuth increments of 90°.

j	B_{0F} [T]	B_{0D} [T]	ϵ_j [rad]	$\alpha_{0,j,F}$ [°]	$\alpha_{0,j,D}$ [°]
1	0.370973	−0.364563	−0.0125	9.359683	16.953433
2	0.314173	−0.330844	−0.1250	31.143486	38.737236
3	0.338564	−0.369984	−0.0125	46.481514	54.075264
4	0.434163	−0.464282	+0.1500	68.265317	75.859067

The remaining parameters are $k = 7.4319$, $\delta = \pi/6$, $\lambda = 0.14$ m and a linear Enge polynomial $(a_0, a_1) = (0, 3.91)$. The aperture parameters are $\Delta r_- = 0.15$ m, $\Delta r_+ = 0.8$ m and $H = 1$ m. Zero-field arcs again complete the reference geometry.

6 Native geometry and orbit checks

The machine figures use vertices stored in native `_ElementPositions.py` files. The footprints are nominal aperture/display volumes, not mechanical yokes or full fringe-field envelopes.

The companion script decodes the compressed vertices without executing the plotting code. It matches every mesh to an input pose and checks the spiral centre and face angles, radial bounds and total arc length. All 65 nominal elements per ring agree with Eqs. (2) and (6); the maximum angular mismatch is below 2×10^{-15} rad. This verifies the geometry conversion, not the accuracy of a tracking integrator.

The existing native COF outputs give the values below. At the $Z = 0$ section, $R_{\mathrm{co}} = R_0 + X_{\mathrm{co}}$ and $Y = P_Y = 0$. Thus R_0 fixes the placement circle, while the magnetic fields determine a different closed orbit.

Table 4: Closed-orbit launch radii and full tunes read from the existing `orbit.json` files in the three machine directories.

Lattice	R_{co} [m]	Q_r	Q_Y
16-cell F/D	4.392925555	3.41066045	3.39111214
55°, D off	4.163263702	3.11774002	1.18920986
Fourfold	3.984968759	3.44738596	3.43884713